

Map, reduce and filter



Agenda

- ① map
- ② reduce
- ③ filter

Higher-order functions

Higher-order function takes function as a parameter and return a function.

① map()

② reduce()

③ filter()

map()

map (function, iterables)

```
def square(a):
```

```
    return a*a
```

1 4 9 16

```
x = map(square, [1, 2, 3, 4])
```

x is a map object

```
l1 = list(x)    or    for e in x:  
                    print(e, end=' ')
```

filter()

filter (function, iterables)

The filter function is used to generate an output list of values that return true when the function is called

```
def f(x):  
    if x >= 3:  
        return x
```

```
y = filter(f, (1, 2, 3, 4))  
l1 = list(y)
```

reduce()

reduce (function, iterables)

The reduce function applies a provided function to iterables and return a single value

```
from functools import reduce
```

```
x=reduce(lambda a,b: a+b, [1,2,3,4])
```

```
print(x)
```

